IEC System for Conformity Testing and Certification of Electrotechnical equipment and Components

CTL PROVISIONAL DECISION SHEET (DSH)

Standard(s) (incl. year)	Subclause(s)	No.	Year
IEC 60065 7th Ed + am1 + am2; IEC 62368-1 2nd Ed IEC 60065 8th Ed. IEC 60950-1 2 Ed. +am1 +am2	15.1.2 (IEC 60065); Annex G.4.3 (IEC 62368-1)	DSH 1099	2014
Category			
OFF / TRON / ITAV			
Subject	**Keywords**	**Developed by**	**To be approved**
Connector Mismatch	- connector mismatch - XLR - IEC 60320 C-7	ETF 2	2015 CTL Plenary Meeting

Question

Is a 3-pin male XLR audio connector (IEC 61076-2-103) allowed to be located immediately adjacent to a C-8 appliance inlet (IEC 60320) in audio equipment?

Decision

No, connectors referenced above are not permitted to be located immediately adjacent to each other in accordance with the present wording of the above subject standards. There is a potential risk of electric shock if the referenced connectors are located immediately adjacent to each other. To help prevent such accidental interconnection, the connectors must be located on separate panels or of sufficient distance away from each other on the same panel so that accidental misconnection is unlikely.

For other consumer applications (for example, home use, and potentially accessible by children), the combination of these connectors shall not be used in the same products as misconnection is foreseeable misuse, potentially resulting in risk of electric shock.

At the 2014 TC 108 meeting in Tokyo this question was discussed and TC 108 agreed with the decision.

Explanatory notes

Issue: The dimensions of the commonly accepted industry standard XLR connector (per IEC 61076-2-103) are such that it would allow for insertion of an IEC 60320 C-7 connector, resulting in a possible risk of electric shock hazard in applications associated with a.c. mains . Pin 1 of the XLR connector, which is typically “chassis ground,” could be accidentally connected to the either the Line or the Neutral of the a.c. mains.

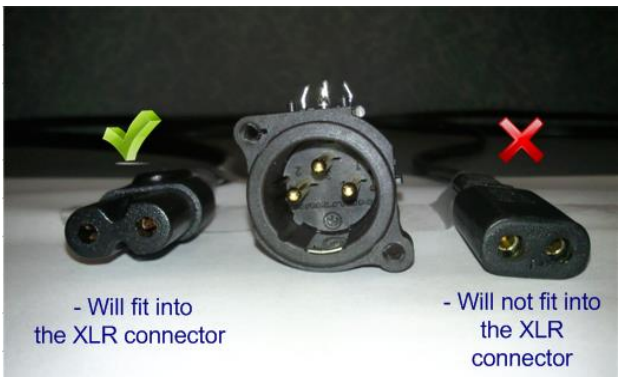
A strict interpretation of the IEC 60065 and 62368-1could imply that products with both types of these connectors would not comply with the requirements of IEC 60065 or IEC 62368-1, especially involving ac mains.

However, in practice, in the commercial or professional audio industry, use of XLR connectors as described is common and acceptable.

For such commercial and professional applications, when the likely user is an “instructed person,” such connectors can be used if the likelihood of accidental interconnection of these connectors is minimized. One such preferred method of minimizing accidental misconnection is locating each

connector on a different panel. Another such method of minimizing accidental interconnection is locating the connectors with sufficient distance away from each other (for example, opposite ends of a panel) so someone (for example, technician) would not accidentally interconnect the connectors by mistake, for example, in low light situations or by feel.

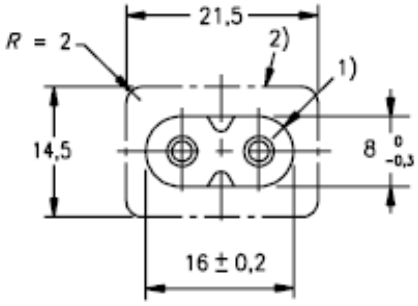
Additional supporting information:

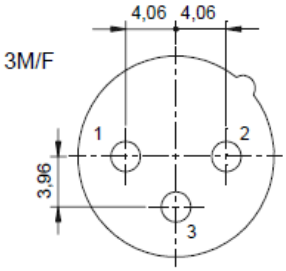




61076-2-103 © IEC:2004

3.3.2 Contact arrangements





© IEC – IECEE 2014